Computer Networks – Course Project

Automated Greenhouse Monitoring and Control System

Role: Team Leader
Inha University, Incheon, South Korea | Mar 2025 – Jun 2025

Project Overview

This project involved designing and implementing an **Automated Greenhouse Monitoring and Control System** using UART and Bluetooth-based communication. The system integrated multiple sensor nodes (Arduino-based) with a centralized laptop server running a Flask web application for real-time monitoring and control.

The architecture supported both **short-range (Bluetooth SPP)** and **long-range (HC-12 UART wireless)** communication, enabling scalable greenhouse section monitoring.

Role and Key Contributions

As **Team Leader**, I was responsible for overall system architecture, communication protocol design, and integration between hardware and software components.

- Designed the **multi-node monitoring architecture**, connecting multiple Arduino-based greenhouse sections to a master laptop node.
- Implemented **UART-based serial communication (9600 bps)** for reliable sensor data transmission via USB and HC-12 modules.
- Configured and tested **Bluetooth SPP (HC-06)** for short-range wireless monitoring through virtual COM ports.
- Integrated the **HC-12 long-range wireless module** (UART-based) to support extended communication distances (up to 1 km).
- Developed a **Flask-based web dashboard** (HTTP server on port 5000) to visualize real-time sensor data.
- Implemented actuator control logic (relay module for water pumps/fans, buzzer alerts) triggered by threshold-based sensor readings.
- Coordinated team development, testing cycles, hardware debugging, and technical documentation.

Technical Challenges and Solutions

1. Multi-Node Communication Synchronization

Challenge:
Simultaneously managing multiple greenhouse nodes over UART and wireless links caused data collisions and inconsistent parsing.

Solution:

- Designed a standardized data packet format with clear delimiters.
- Implemented sequential polling from the master node.
- Added buffer handling and parsing validation to prevent corrupted readings.

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2. Wireless Communication Stability

Challenge:
Bluetooth disconnections and interference affected reliability.

Solution:

- Implemented reconnection handling logic.
- Optimized baud rate settings and reduced packet size for stable transmission.
- Used HC-12 for long-range communication to improve robustness.

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3. Real-Time Monitoring via Web Interface

Challenge:
Ensuring timely updates of sensor data without blocking serial communication.

Solution:

- Structured Flask routes for efficient data handling.
- Separated serial data acquisition from HTTP request handling logic.
- Implemented color-coded UI indicators for threshold-based alerts.

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4. Hardware-Software Integration

Challenge:
Coordinating sensor readings with actuator control (e.g., watering, buzzer alerts) without false triggering.

Solution:

- Defined threshold-based control logic.
- Added validation checks before triggering relays or alarms.
- Logged recent commands (24-hour history) for traceability.

Outcome

The final system successfully demonstrated:

- Reliable **UART and wireless serial communication**
- Real-time **multi-node environmental monitoring**
- Remote visualization through a **Flask-based web dashboard**
- Automated actuator control based on environmental conditions

The project strengthened practical skills in **network protocols, embedded systems integration, wireless communication, and client-server architecture**.